Stochastic Modelling of Soil Nitrogen in Intensive Agriculture

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Abstract

This dissertation proposes, justifies and evaluates a system of stochastic differential equations to model soil nitrogen in the context of dairy farming. Special focus is placed on the stochastic dynamics of high nitrogen concentration urine patches and the resulting effects on nitrogen leaching rates. The presented model considers the evolution over time of soil nitrogen concentration, pasture density and soil water storage at a single point. The random processes of urination events, rainfall events and pasture consumption events are described by compound Poisson processes. Model realisations are sampled using numerical methods, with Monte Carlo simulations being used to find typical results for given model parameters.

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Chapter 1

Introduction

1.1 Motivation

Excess nitrogen in waterways can contaminate drinking water and cause excessive algae growth. In cases of drinking water highly contaminated with nitrates, infants can potentially develop blue baby syndrome [23]. Livestock urine is the largest source of nitrogen in the waterways of New Zealand, with dairy farms in particular being significant contributors [15]. Soil nitrogen concentrations in dairy farms are characterised by urination events with high nitrogen concentrations. Due to the high concentration in small patches, pasture growth and denitrification processes can not remove all the deposited nitrogen, causing excess nitrogen to leach into groundwater and waterways. Finding methods to effectively model nitrogen leaching rates as a function of farming methods would be an invaluable tool to aid environmentally conscious farming decisions.

The location, timing and intensity of livestock urination events are stochastic in nature, leading to interesting dynamics that are not otherwise considered in conventional nitrogen cycle models. There is a need to develop stochastic models which capture the unique dynamics of nitrogen concentrations, and thus nitrogen leaching, in the context of dairy farms.

1.2 Related Works

The general topic of nitrogen cycle modelling is well studied, with many models of varying complexity having been developed. Notable models include the Daisy model [19], the DNDC model [27], and the APSIM model [22]. All three models are comprehensive, with many constituent sub-models considering a wide range of soil-plant-atmosphere dynamics. The Daisy model was developed with Northern European soils and climates in mind, and focuses on nitrogen cycling. The DNDC model was developed in the United States to estimate greenhouse gas emissions and during later developments it has maintained a focus on emissions which

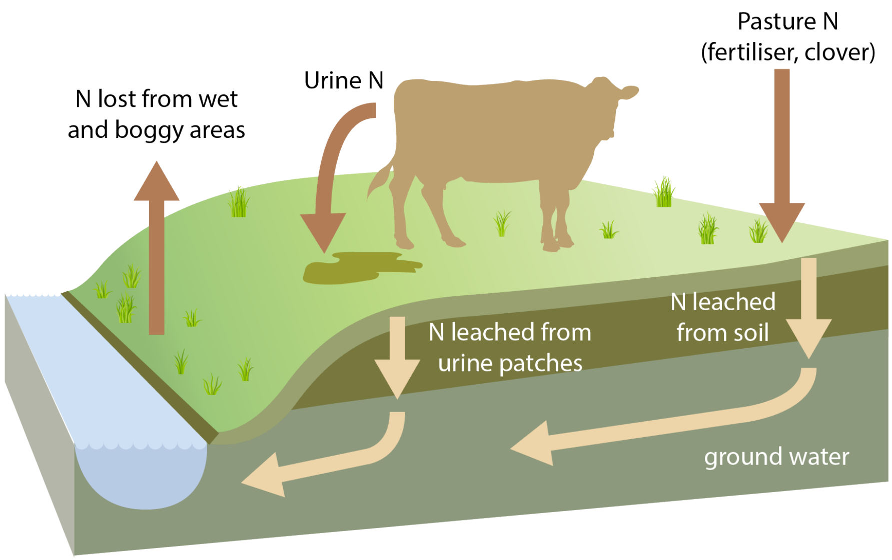


Figure 1.1: Simplified nitrogen cycle in dairy farms. Source: [37]

has lead to detailed modelling of microbile soil processes. A variation of the DNDC model called NZ-DNDC was developed with soil and livestock inputs specific to New Zealand [16]. The Daisy model does not consider livestock urine inputs whereas the DNDC, NZ-DNDC and APSIM models do. However, urine inputs are generally modelled in a dertiministic manner in these models with stochastic dynamics being ommited.

Although there exists significant research regarding nitrogen modelling in general, much less work has been done to consider the stochastic dynamics of urine deposits. Work produced by Taylor et al. presents a patch model which considers the concentration of nitrogen due to overlapping urine patches [35, 36]. A density function $\phi(x,t)$ representing the distribution of nitrogen concentrations of the patch model is introduced. The evolution of $\phi(x,t)$ is described by a partial differential equation (PDE) and considers discrete jumps in nitrogen concentrations following a urination event. The evolution PDE of $\phi(x,t)$ in a toy case is written by Taylor et al. as,

$$\frac{\partial}{\partial t}\phi(x,t) = \beta\frac{\partial}{\partial x}x\phi(x,t) + \lambda\phi(x-\alpha,t) - \lambda\phi(x,t) \tag{1.1}$$

where x is the nitrogen concentration, t is time, α is the change in nitrogen concentration following a urination event, β is the rate of exponential decay, and λ is the rate of urination events.

A widely used model within the New Zealand agricultural industry is the Overseer model [30]. Overseer covers many aspects of nutrient modelling, including urination dynamics, and is used to inform water quality regulations by some regional councils. Despite the complexity of the model, a 2018 review panel from the New Zealand Parliamentary Commissioner for the Environment stated that they do "not have confidence that Overseer's modelled outputs tell us whether changes in farm management reduce or increase the losses of nutrients, or what the magnitude or error of these losses might be" [31]. The panel cited several potential shortcomings with the model:

- A focus on just nitrate nitrogen.
- A lack of focus on overland flow, a significant pathway of nutrient loss during high rainfall.
- As Overseer uses an "average climate", it is not able to capture discrete weather events.
- A single soil layer is modelled, and uniform water movement is assumed.
- Inputs and outputs of nitrogen are not balanced; thus, the model is not constrained by any fundamental conservation laws.

Many details of the Overseer model are not publically available in their current form. As a result, despite the close alignment of Overseer to the topic of this dissertation, none of the sub-models or details of Overseer as used in the remainder of this dissertation.

Chapter 2

Preliminaries

Definition 2.0.1 (Poisson distribution)**.** A discrete random variable N is said to have a Poisson distribution with parameter $\lambda > 0$ if it has the following probability mass function (PMF),

$$f(n) = \text{Pr}(N = n) = \frac{\lambda^n e^{-\lambda}}{n!}$$

Where n is the number of event occurrences ($n = 0, 1, 2, \ldots$). The variance and expectation of the Poisson distributed random variable N are given by,

$$\text{E}(N) = \text{Var}(N) = \lambda$$

If N is Poisson distributed we write $N \sim \text{Pois}(\lambda)$.

Definition 2.0.2 (Exponential distribution)**.** A continuous random variable X is said to have an exponential distribution with parameter $\lambda > 0$ if it has the following probability density function (PDF),

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

The variance and expectation of the exponentially distributed variable X are given by,

$$\text{E}(X) = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

If X is exponentially distributed we write $X \sim \text{Exp}(\lambda)$.

Definition 2.0.3 (Normal distribution)**.** A continuous random variable X is said to have an normal distribution with parameters μ and σ^2 if it has the following PDF,

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The variance and expectation of the normally distributed variable X are given by,

$$\mathrm{E}(X) = \mu$$

$$\mathrm{Var}(X) = \sigma^2$$

If X is normally distributed we write $X \sim \mathcal{N}(\mu, \sigma^2)$.

Definition 2.0.4 (Log-normal distribution)**.** A continuous random variable X is said to have a log-normal distribution with parameters μ and σ^2 if and only if $\ln(X) \sim \mathcal{N}(\mu, \sigma^2)$.

Definition 2.0.5 (Homogeneous Poisson point process)**.** A homogeneous Poisson point process (PPP) is a random process with counting variable $N(t)$ and rate λ defined by the following properties for $t \geq 0$:

1. $N(0) = 0$.
2. The number of events over an interval of length t is distributed as $N(t) \sim \mathrm{Pois}(\lambda t)$.
3. The distribution of events of a PPP in disjoint time intervals are independent of one another.

Definition 2.0.6 (Compound Poisson process)**.** A compound Poisson process (CPP) of rate λ is a continuous-time random process given by,

$$Y(t) = \sum_{i=1}^{N(t)} Z_i$$

where $N(t)$ is the counting variable of a PPP of rate λ and $\{Z_i, i \geq 0\}$ are independent, identically distributed random variables with probability density function $g(z)$ (also called jump size distribution).

Remark. For the remainder of this dissertation, PPP can be assumed to mean a homogeneous PPP. We will also assume that the PPP defining a CPP is a homogeneous PPP.

Remark. Note that at times during this dissertation, the independent variable of a PPP $N(t)$ or a CPP $Y(t)$ may be subscripted for convenience, giving N_t and Y_t respectively.

Chapter 3

Random Jump Process Model

3.1 Proposed Model

In contrast with the previous patch model produced by Taylor et al., [36], we will now consider the modelling of nitrogen concentrations at a singular point in a given paddock (rather than the paddock as a whole). Let us label this point as P for future reference. The evolution of nitrogen concentration at point P is influenced by several variables such as water concentration, pasture density, fertiliser usage, effluent irrigation... etc. [13, 37]. In addition, nitrogen can exist in several forms within the soil, including ammonia, nitrate, and nitrous oxide, and can transform from one form to another [33]. For the purposes of this dissertation we simplify our model to only consider total nitrogen concentration (an accumulation of all forms), soil water storage and pasture density within the topsoil at point P. We define topsoil as the top 300 mm of soil. Time units are measured in days unless otherwise specified. Let us introduce the following random processes, all of which model their respective quantities at point P,

- $X_N(t)$ is the total nitrogen concentration at time t with units kg/ha .
- $X_W(t)$ is the soil water storage at time t with units mm.
- $X_P(t)$ is the dry mass (DM) pasture density at time t with units kg/ha.

Let us compactly express these three random variables as the vector,

$$\mathbf{X}_t = [X_N(t), X_W(t), X_P(t)]^\intercal \tag{3.1}$$

Similar to Taylor et al., we wish to construct a model that assumes nitrogen concentrations take a discontinuous jump upon a urination event. We will justify the choice of a CPP as a model for these jumps in Section 3.3.1. Dzupire et al. suggest that a CCP is an appropriate simplistic model for rainfall events [12]. We will assume that the consumption of pasture at a particular point by livestock also follows a CPP (although it should be noted that there is limited existing research in this area to justify this claim). Let us define the following CPPs to model these phenomena at point P,

- $Y_N(t)$ is the accumulation of nitrogen concentration due to livestock urination events at time t.
- $Y_W(t)$ is the soil water storage accumulation from rain events at time t.
- $Y_P(t)$ is the accumulated portion of pasture consumed by livestock at time t.

$Y_N(t)$, $Y_W(t)$ and $Y_P(t)$ have Poisson rates of λ_N, λ_W and λ_P, and jump size distributions of $g_N(z)$, $g_W(z)$ and $g_P(z)$ respectively. Let us introduce the following vector notion to express these random processes at time t,

$$\mathbf{Y}_t = [Y_N(t), Y_W(t), Y_P(t)]^\intercal \tag{3.2}$$

In addition to stochastic changes in $\mathbf{X}_t$ modelled by CPPs, deterministic drifts modelled by differential equations must also be considered. Naturally, this points us towards a system of stochastic differential equations (SDEs) driven by CPPs as an appropriate model. We therefore propose the following system of SDEs,

$$\begin{bmatrix} dX_N(t) \\ dX_W(t) \\ dX_P(t) \end{bmatrix} = \begin{bmatrix} f_N(\mathbf{X}_t, t)dt \\ f_W(\mathbf{X}_t, t)dt \\ f_P(\mathbf{X}_t, t)dt \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & X_P(t) \end{bmatrix} \begin{bmatrix} dY_N(t) \\ dY_W(t) \\ dY_P(t) \end{bmatrix}$$

or more compactly,

$$d\mathbf{X}_t = \mathbf{f}(\mathbf{X}_t, t)dt + \mathbf{J}(\mathbf{X}_{t^-}, t^-)d\mathbf{Y}_t \tag{3.3}$$

where,

- $\mathbf{X}_t \in \mathbb{R}^3$ is a realisation of the random process at time t with the components defined in Eq. (3.1).
- $\mathbf{Y}_t \in \mathbb{R}^3$ is a realisation of the vector of CPPs at time t as defined in Eq. (3.2).
- $\mathbf{f}(\mathbf{X}_t, t) = [f_N(\mathbf{X}_t, t), f_W(\mathbf{X}_t, t), f_P(\mathbf{X}_t, t)]^\intercal \in \mathbb{R}^3$ is the drift vector representing deterministic changes to $\mathbf{X}_t$.
- $\mathbf{J}(\mathbf{X}_{t^-}, t^-) \in \mathbb{R}^3 \times \mathbb{R}^3$ is the matrix mapping the changes in each element of $\mathbf{Y}_t$ to each element of $\mathbf{X}_t$. For future reference, let us name the rows of $\mathbf{J}$ as $\mathbf{J}_1$, $\mathbf{J}_2$ and $\mathbf{J}_3$. The choice of $\mathbf{J}$ is justified in Section 3.3.

For the remainder of this dissertation, the term *the model* refers to Eq. (3.3) and results of *the model* refer to the realisations or the aggregation of realisations of Eq. (3.3), numerically simulated or otherwise.

3.2 Drift Term Modelling

Having introduced the general form of *the model* in Eq. (3.3) let us now consider the details of each term, beginning with the drift term $\mathbf{f}(\mathbf{X}_t, t)$. Modelling the deterministic drift in $\mathbf{X}_t$ can be approached with varying degrees of complexity. As *the model* does not have any spatial dimensions, a significant source of complexity has already been eliminated. A limitation due to the simplicity of *the model* is that some complex processes in the nitrogen cycle can not be completely captured with a single metric for nitrogen concentration or pasture density. Many existing pasture growth and nitrogen models differentiate between various nitrogen compounds and between above-ground plant matter and below-ground plant matter [19, 32]. With these limitations in mind, let us consider each element of $\mathbf{f}(\mathbf{X}_t, t)$.

3.2.1 Nitrogen Concentration Drift

We will break down the influences on nitrogen concentration changes into livestock excrement, input from fertilisation, pasture uptake of nitrogen, nitrogen leaching and denitrification. Livestock excrement modelling is captured by the jump term of *the model*. Let us write,

$$f_N(\mathbf{X}_t, t) = I_N(t) - U_N(X_N, X_W, X_P) - L_N(X_N, X_W) - D_N(N)$$

where,

- $I_N(t)$ is the input from fertilisation.
- $U_N(X_N, X_W, X_P)$ is the rate of pasture uptake of nitrogen.
- $L_N(X_N, X_W)$ the rate of nitrogen leaching.
- $D_N(X_N)$ is the rate of nitrogen denitrification.

Input of nitrogen from fertilisation depends on the methods and practices of a given farm and may vary in quantity and frequency. A common frequency of fertiliser application is four times per year. We will assume the fertiliser is applied in four even applications per year (i.e. every $365/4$ days) with a total annual quantity of β_F. Assuming that there are discrete application events of quantity $\beta_F/4$ kg/ha of nitrogen we can write $I_N(t)$ in terms of the Dirac delta function δ as,

$$I_N(t) = \frac{\beta_F}{4} \sum_{k=1}^{\infty} \delta(t - \frac{365}{4}k)$$

Pasture uptake of nitrogen shows a strong linear correlation to pasture growth [14]. Therefore, we will write pasture uptake of nitrogen as a scalar multiple of a pasture growth function

$G_P(X_N, X_P, X_W)$ (defined in Section 3.2.3). If we introduce β_U as the coefficient scaling pasture growth to pasture uptake of nitrogen, we get,

$$U_N(X_N, X_W, X_P) = \beta_U G_P(X_N, X_W, X_P)$$

Empirical results from [11] suggest that the rate of nitrogen leaching per drainage volume has approximately a cubic relationship with nitrogen concentration. With β_L being the rate of leaching coefficient and $D_W(X_W)$ being the drainage rate of water (defined in Section 3.2.2), let us approximate nitrogen leaching as such,

$$L_N(X_N, X_W) = \beta_L X_N^2 D_W(X_W)$$

We should now note that L_N is responsible for the leaching of nitrogen into waterways, a key motivation for this research. Paying closer attention to this term we can see that total nitrogen leached L_{total} during some time interval $[t_0, t_1]$, with units kg/ha, is the integral,

$$L_{total}([t_0, t_1]) = \int_{t_0}^{t_1} L_N \, dt$$

For a given interval $[t_0, t_1]$, let us also define the term L_{rate} as the mean rate of nitrogen leaching over that interval with units kg/ha/year. L_{rate} derived from a large time interval will allow to start developing an estimate of typical leaching rates for a given farm. We simply define L_{rate} as such,

$$L_{rate}([t_0, t_1]) = \frac{365 L_{total}([t_0, t_1])}{t_1 - t_0}$$

Denitrification is a complicated process which is influenced by many variables, including temperature, soil acidity, oxygen availability, water concentration and nitrogen concentration (more specifically nitrate concentrations) [5, 11]. Given the complexity of this process within the constraints of *the model*, we will make the crude assumption that denitrification is modelled by exponential decay of nitrogen concentrations at rate $\beta_D X_W$,

$$D_N(X_N, X_W) = \beta_D X_N X_W$$

3.2.2 Soil Water Storage Drift

We will break down the influences on soil water storage into rainfall, drainage and evaporation. Given that rainfall is modelled by the jump term of (3.3), we write,

$$f_W(\mathbf{X}_t, t) = -D_W(X_W) - E_W(t)$$

where

- $D_W(X_W)$ is the rate of water loss due to drainage.

- $E_W(t)$ is the rate of evaporation.

Research by Black et al. shows that the rate of drainage occurs at an exponential rate relative to soil water storage [3]. For a drainage constants α_W and β_W we write,

$$D_W(X_W) = \alpha_W e^{\beta_W X_W}$$

Water evaporation is dependent on several variables, including temperature, humidity, wind speed, soil properties and vegetation cover. We will make the crude assumption that evaporation rates are constant at rate β_E,

$$E_W(t) = \beta_E$$

3.2.3 Pasture Density Drift

Finally, we will consider the factors influencing changes in pasture density. We will consider just natural pasture growth and animal grazing consumption of pasture. Animal grazing will be modelled by the jump term of (3.3). Thus, we can simply write,

$$f_P(\mathbf{X}_t, t) = G_P(X_N, X_W, X_P)$$

where $G_P(X_N, X_W, X_P)$ is the rate of pasture natural growth.

Research by D'aoust et al. suggests that grass growth rates are dependent on both water concentration and nitrogen concentration [8]. We will approximate pasture growth with logistic growth, proportional to soil water storage and nitrogen concentration. We define parameters β_G and β_M as the rate of pasture growth and the maximum pasture density, respectively, giving,

$$G_P(X_N, X_W, X_P) = \beta_G X_N X_W X_P \left(1 - \frac{X_P}{\beta_M}\right)$$

3.3 Jump Term Modelling

We now consider the appropriate details for the jump term of *the model* defined in Eq. (3.3). As defined in Section 3.1, the jump term $\mathbf{Y}_t$ is a vector of three CPPs, describing jumps attributed to urination events, rainfall events and pasture consumption events, respectively. Let us consider each element individually.

3.3.1 Urination Events

There are several connected assumptions that must be made to describe the jump in nitrogen concentration following a urination event.

We will first justify that the urination events at a particular point in a paddock are, in fact, accurately modelled by a PPP. We begin by assuming that there is a constant rate of urination events produced by a herd of cows. We assume that our herd is of sufficient size such that by the law of large numbers, any variance in the rate of urination events of individual cows is negligible on the scale of a herd. In reality, a constant rate may not be completely accurate as the whole herb may have lower or higher rates of urination events at different times of the day (i.e. night versus day). Let this rate of urination events for the whole herd be γ. That is, over a given time interval $[t_0, t_1]$ where $t_0 < t_1$, we can approximate the total number of urination events n as,

$$n \approx \text{round}((t_1 - t_0)\gamma)$$

We can think of the number of urine patches hitting point P over this time interval $[t_0, t_1]$ as the summation of n Bernoulli samples. Each Bernoulli sample represents the event that a given urine event produces a urine patch that hits point P. Betteridge et al. found that the distribution of the volume of cattle urine across urination events closely follows a log-normal distribution [1]. We assume that the urine in a given patch is even distributed across the patch. Therefore, the area of a urine patch will vary between urination events and is also distributed by a log-normal distribution. We assume that urine patches are uniformly distributed throughout the paddock. Therefore we define,

- $A_1, A_2, \ldots, A_n$ are independent, identically distributed random variables following a log-normal distribution. A_i represents the area of urine patch i.
- $B_1, B_2, \ldots, B_n$ are independent random variables, each with a Bernoulli distribution (not necessarily sampled from the same distribution). B_i represents the outcome of the event that point P is hit by urine patch i.
- $\Pr(B_i = 1) = p_i$ for $i = 1, 2, \ldots, n$. Given our assumption of uniformly distributed urine patch locations, the probability of point P being hit by patch i is $p_i = A_i/A_{paddock}$ where $A_{paddock}$ is the area of the paddock (i.e. hitting probability is proportional to patch area).
- $\lambda_n = p_1 + p_2 + \cdots + p_n = (A_1 + A_2 + \cdots + A_n)/A_{paddock}$.
- $N_n = B_1 + B_2 + \cdots + B_n$ is the number of urine patches hitting our given point of interest. By definition, N_n follows a Poisson binomial distribution.

Theorem 3.3.1 (Le Cam, [24])**.** *For a random variable N following a Poisson binomial distribution with Bernoulli distributions $q_1, q_2, \ldots, q_n$., and for $\lambda = q_1 + q_2 + \cdots + q_n$ Le Cam's theorem for Poisson binomial distributions states,*

$$\sum_{k=0}^{\infty} \left| \Pr(N = k) - \frac{\lambda^k e^{-\lambda}}{k!} \right| < 2 \left(\sum_{i=1}^{n} q_i^2 \right) \tag{3.4}$$

Thus, by Le Cam's theorem, N_n is approximately Poisson distributed with the bounds of the approximation error described by the inequality Eq. (3.4).

Let us try to quantify the approximation error of Eq. (3.4) using the upper bounds of typical diary farm parameters. The typical area of cow urine patches is well less than 1 m^2 [20] and a typical rotational grazing dairy paddock is greater than 1 ha (or 10,000 m^2) in area for a standard sized herd [18]. Therefore $p_i < 10^{-4}$ for almost every urination event i given these bounds and thus $p_i^2 < 10^{-8}$. Furthermore, the rate of urination events is less than 20 per cow per day [34], and 94.8% of New Zealand dairy herds have less than 1000 cows [28]. Thus, over the time interval of 1 day, we conclude that $n <$ 20,000 urination events for almost every herd. Therefore

$$\sum_{k=0}^{\infty} \left| \Pr(N_n = k) - \frac{\lambda_n^k e^{-\lambda_n}}{k!} \right| < 4 \times 10^{-4}$$

An approximation error bound of 4×10^{-4} is negligible for the purposes of our modelling and will only become smaller as we consider smaller time intervals or less extreme diary farm parameter bounds. Thus, we conclude that,

$$N_n \sim \text{Pois}(\lambda_n)$$

Now, we show that the variance of λ_n is sufficiently small for it to be considered constant for a given time interval (and thus for a given n). Once again, we will use the bounds of typical diary farm parameters to justify this assumption. [20] found that cow urine patches have a mean area of approximately 0.5 m^2 with a standard deviation of approximately 0.15 m^2. Therefore

$$\text{E}(A_i) = 0.5, \quad \text{Var}(A_i) = 0.15^2, \quad i = 1, 2, \ldots, n$$

and thus

$$\text{E}(\lambda_n) = \frac{0.5n}{A_{paddock}}, \quad \text{Var}(\lambda_n) = \frac{0.15^2 n}{A_{paddock}^2}$$

Therefore, λ_n has a standard deviation of

$$\text{SD}(\lambda_n) = \frac{0.15\sqrt{n}}{A_{paddock}}$$

If we rewrite the standard deviation in terms of the expected value of λ_n, we find

$$\text{SD}(\lambda_n) = \frac{0.3}{\sqrt{n}} \text{E}(\lambda_n)$$

Let us find a lower bound for n. The rate of urination events is greater than 5 per cow per day [34] and 98.9% of New Zealand dairy herds have more than 100 cows [28]. Thus, over the time interval of one day, we conclude that $n > 500$ urination events. Therefore we can bound the standard deviation of λ_n as

$$\mathrm{SD}(\lambda_n) < \frac{0.3}{\sqrt{500}}\mathrm{E}(\lambda_n)$$

This equates to the standard deviation of λ_n being less than approximately 1.3% of the expected value of λ_n. This standard deviation bound will only tighten given less extreme parameter bounds. Therefore, we can conclude that the variance in λ_n is insignificant compared to its magnitude, and thus, for a given length of time period, we can treat λ_n as approximately constant.

We have shown for a given time interval $[t_1, t_2]$ that N_n follows a Poisson distribution of rate λ_n. In addition, the number of urination events at a given point over two non-overlapping intervals $[t_1, t_2]$ and $[t_3, t_4]$ both follow independent Poisson distributions given our independence assumptions. We found that λ_n is approximately constant for a given herd and paddock and is given as

$$\begin{aligned}\lambda_n &\approx \frac{nE(A_i)}{A_{paddock}} \\ &\approx \frac{(t_1 - t_0)\gamma E(A_i)}{A_{paddock}}\end{aligned}$$

Let us define $\lambda_N = \frac{\gamma E(A_i)}{A_{paddock}}$ and let $\Delta t = t_1 - t_0$. Note that λ is constant for a given herd and paddock. Therefore we have

$$\lambda_n = \Delta t \lambda_N$$

and thus, as the rate λ_n is proportional to time and independence properties have been satisfied, we can conclude that the occurrence of urination events at a given point in a paddock in continuous time t are modelled by a homogeneous PPP $N_N(t)$ of rate λ_N. It should be noted that this was only justified using time intervals in the order of one day. If we consider very short time intervals then some of our justification breaks down. For example, when considering time in the magnitude of seconds or minutes (rather than several hours or days), then the rate of urination events in the paddock no longer appears continuous, and instead, discrete urination events occur with potentially varied wait times between events. This breaks our assumption that the urination events in a paddock occur at a continuous rate γ. With this being said, the issue of nitrogen pollution on dairy farms does not require minute-on-minute observations or modelling and is more of a macro time scale problem, thus justifying our use of larger time scales for our assumptions.

Given we have established that urination events at point P follow a PPP, let us now find an appropriate jump size distribution of changes in nitrogen concentration following a urination event. That is, let us find an appropriate jump size distribution $g_N(z)$ for the CPP $Y_N(t)$. It is observed in [1] that nitrogen concentration (kg/L) in cattle urine follows a log-normal distribution. As we assume that urine is evenly distributed within a given urine patch, it follows that the increase in nitrogen concentration (kg/ha) at point P following a urination event would also be log-normally distributed. Therefore, we let $g_N(z)$ be the log-normal distribution probability density function with parameters μ_N and σ_N.

If we assume that all the nitrogen from a urination event becomes soil nitrogen and no nitrogen contribution from rainfall events or pasture consumption events, the first row of $\mathbf{J}(\mathbf{X}_{t^-}, t^-)$ is,

$$\mathbf{J}_1(\mathbf{X}_{t^-}, t^-) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

3.3.2 Rainfall Events

Dzupire et al. suggest that a stochastic model for rainfall can be achieved by considering rainfall as a series of discrete events, each with a rainfall quantity sampled from a probability distribution [12]. A CPP with an exponential jump size distribution is suggested as a simple model. Adopting this model, we will let the CPP $Y_W(t)$ of rate λ_W follow the jump distribution $g_W(z)$ where $g_W(z)$ is the exponential distribution PDF with parameter θ_W. It should be noted that *the model* does not distinguish between water absorbed by the soil and water yet to be absorbed. We can alternatively think of this simplification as assuming that soil can instantly adsorb an arbitrarily large quantity of rainfall. Thus, $X_W(t)$ experiences jumps in value equal to the jumps in $Y_W(t)$ as all rainfall water is assumed to become soil water storage. We will assume that the only jump term influencing $X_W(t)$ is $Y_W(t)$. It could be argued that the soil water storage could also increase following urination events modelled by $Y_N(t)$; however, we consider this effect negligible. Therefore we can write the second row of $\mathbf{J}(\mathbf{X}_{t^-}, t^-)$ as,

$$\mathbf{J}_2(\mathbf{X}_{t^-}, t^-) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

3.3.3 Pasture Consumption Events

We assume that pasture consumption events by livestock at point P follow a PPP of rate λ_P. Similar arguments as used in Section 3.3.1 could potentially be used to justify the PPP as an appropriate model; however, there is little existing research to quantify exactly what portion of the pasture at a particular point in a paddock is consumed by livestock.

To reflect the nature that livestock consume larger quantities of dense pasture than sparse pasture during a consumption event, we define the third row of $\mathbf{J}(\mathbf{X}_{t^-}, t^-)$ as,

$$\mathbf{J}_3(\mathbf{X}_{t^-}, t^-) = \begin{bmatrix} 0 & 0 & -X_P(t) \end{bmatrix}$$

It follows that the jumps in $Y_P(t)$ are the portions of pasture at point P being consumed each consumption event. Without existing research being able to point us towards a particular distribution from which the size of consumption proportions comes, we settle with the delta distribution. That is, we write the jump size distribution of $Y_P(t)$ as $g_P(z) = \delta(z - \beta_C)$ for the parameter $0 < \beta_C < 1$.

3.4 Model Simulations

We have now defined the *the model*, the system of SDEs as defined in Eq. (3.3), so let us numerically simulate many realisations of our system to generate results in a Monte Carlo style. Given the complexity of *the model*, analytical results are not possible, leading to a Monte Carlo simulation as a viable alternative to generate results. The numerical system used is discussed in Section 4.

3.4.1 Method

We will focus our attention on a particular subset of case studies rather than attempting to run Monte Carlo simulations for every possible profile of a dairy farm. We will consider the Waikato region of New Zealand due to its large dairy industry and significant body of literature. The literature will be consulted to determine the appropriate parameters for each process defined in sections 3.2 and 3.3. It should be noted that while some research exists allowing us to select parameters specific to the Waikato region, many parameters used in the following simulations are derived from general agricultural research. Furthermore, some processes (such as denitrification rates) pose difficulties in finding exact parameters due to their complex nature. Therefore, several parameters are derived using crude assumptions or estimations, with the assistance of reference material (i.e. these parameter values are not explicitly stated in reference material). More detailed field research to better quantify the process parameters such as denitrification, is a point of future research.

We will allow two parameters to vary in our investigations. Namely, we will investigate the impact of changes in stocking rate s and nitrogen fertilisation rate β_F on the components of $\mathbf{X}_t$ and the rate of nitrogen leaching L_{rate} over our simulation interval. These two parameters in particular are selected to vary, as it is within the control of a farmer to change stocking rate and fertilisation input to potentially reduce nitrogen leaching, whereas the characteristics of the soil and rainfall are (to a certain extent) fixed for a given farm. As a result, some parameters are written in terms of s; for example, the Poisson rate of pasture consumption events λ_P is dependent on the stocking rate s.

A simulation time interval of 365 days is used ($t \in [0, 365]$), and 10,000 realisations of $\mathbf{X}_t$ are sampled for each pair of s and β_F values. Values of $\mathbf{X}_t$ and L_{rate} are recorded for each realisation. Values of s and β_F are tested in a 'grid search' style; that is, simulations are run for every combination (s, β_F) in our parameter space $S \times B_F$. We consider the following parameter space,

$$S = \{0, 1, 2, 3, 4, 5\}$$

Parameter	Description	Value	Reference
$X_N(0)$	Nitrogen concentration initial condition.	1×10^2	
$X_W(0)$	Soil water storage initial condition.	2.5×10^1	
$X_P(0)$	Pasture density initial condition.	2×10^3	
λ_N	Poisson rate of urination events.	$5s \times 10^{-4}$	[20]
λ_W	Poisson rate of rainfall events.	1×10^0	[29]
λ_P	Poisson rate of pasture consumption events.	$s \times 10^{-1}$	[7]
μ_N	Log-normal urine nitrogen concentration mean.	5.62×10^0	[2]
σ_N	Log-normal urine nitrogen concentration SD.	7.91×10^{-1}	[2]
θ_W	Rainfall event exponential distribution parameter.	3×10^0	[29]
β_D	Denitrification decay rate.	1.5×10^{-4}	[4]
β_C	Consumption portion per consumption event.	1×10^{-1}	[7]
β_G	Pasture growth rate.	1×10^{-5}	[8]
β_L	Rate of nitrogen leaching.	1.43×10^{-6}	[11]
β_M	Maximum pasture density.	3.8×10^3	[6]
β_U	Nitrogen uptake as a portion of pasture growth.	3×10^{-2}	
β_W	Water drainage parameter.	3.5×10^{-1}	[3]
α_W	Water drainage parameter.	9.63×10^{-5}	[3]

Table 3.1: Parameters used during simulations.

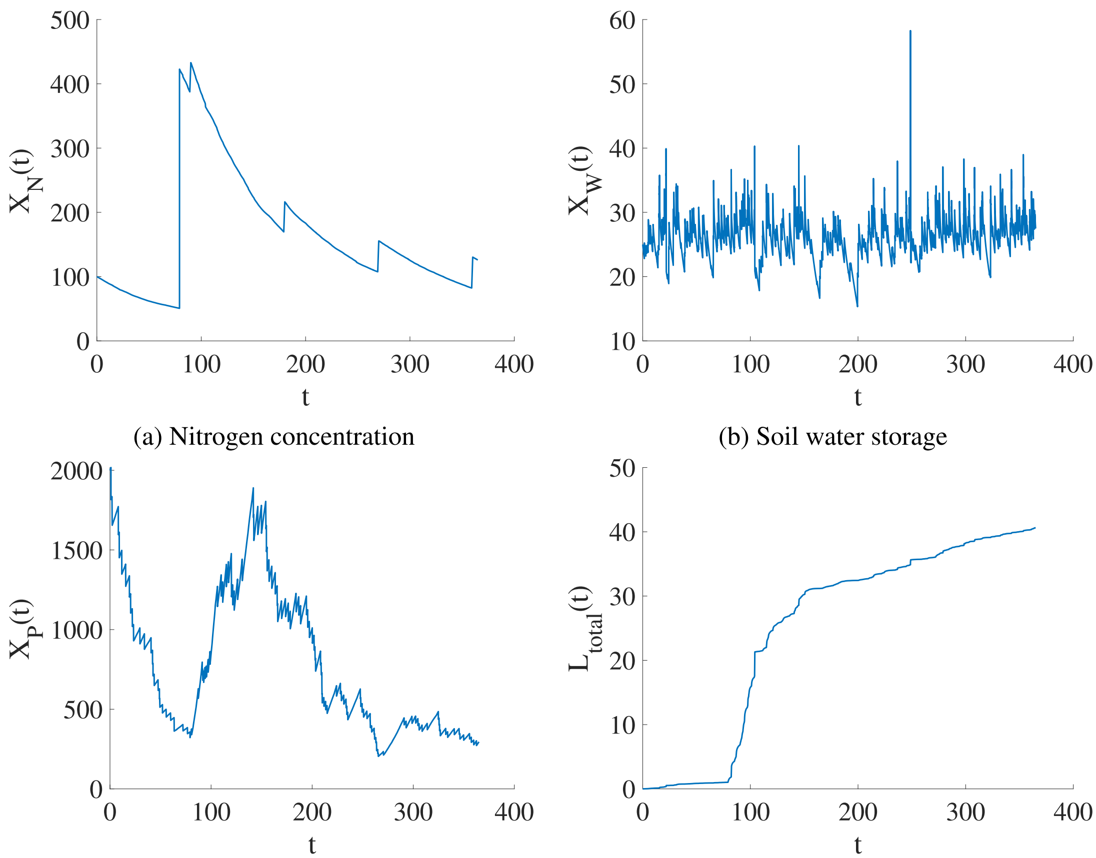
and,

$$B_F = \{0, 66.7, 133.3, 200, 266.7, 333.3, 400\}$$

A stocking rate of $s = 0$ is included in S as a control. Values included in S and B_F are chosen as realistic ranges of stocking rates and nitrogen fertilisation rates. Six stocking rates and seven nitrogen fertilisation rates are trialled; thus, there are 42 combinations in the parameter space. Given that 10,000 realisations are sampled for each parameter combination, the total compute time becomes rather large if we increase the parameter space any more. Furthermore, we find that small changes in s and β_F result in small changes in $\mathbf{X}_t$ and L_{rate}; therefore, we do not lose significant information from using a coarse parameter space.

3.4.2 Simulation Results

In Figure 3.1 we see an example of a single realisation of $\mathbf{X}_t$ for $s = 3$ and $\beta_F = 200$. The changes in nitrogen concentration $X_N(t)$ see a large jump from a urination event at $t \approx 80$ alongside small quarterly jumps from fertiliser application. We see the expected behaviour of pasture growth following a urination event and also pasture density declines in times of low nitrogen concentration. As expected, there is a large increase in the accumulated nitrogen leaching L_{total} following the urination event.



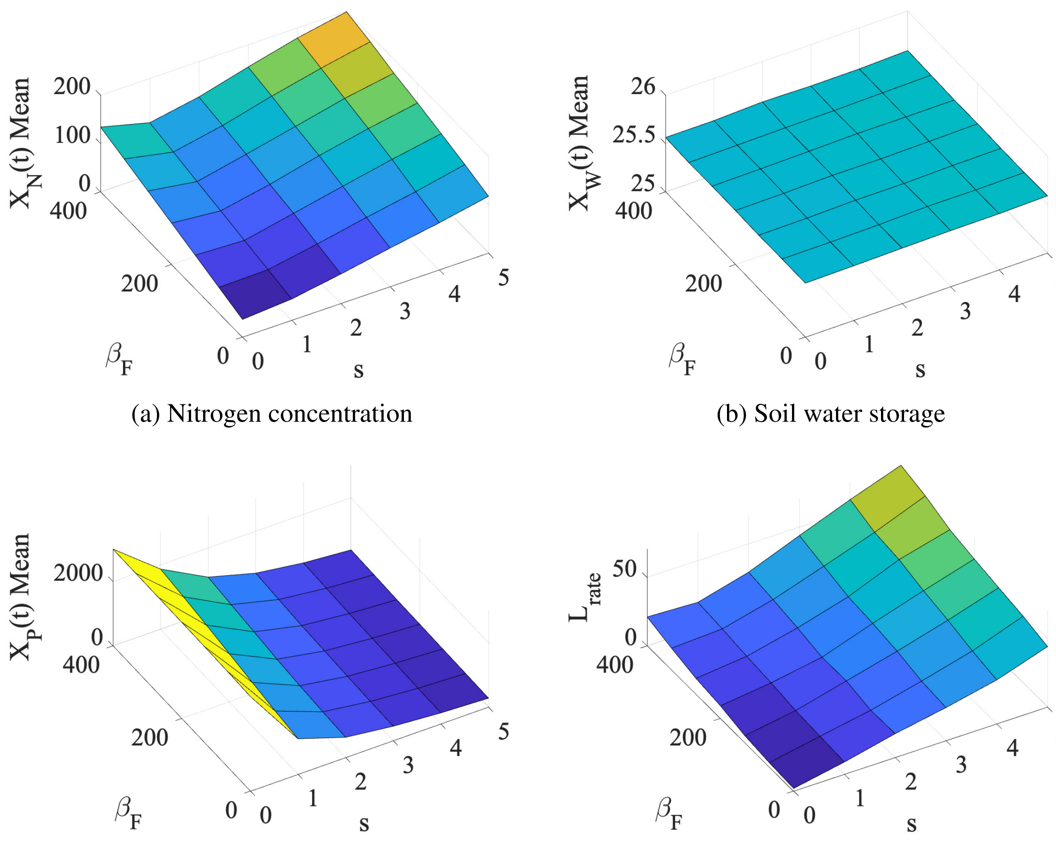
(a) Nitrogen concentration (b) Soil water storage

(c) Pasture density (d) Nitrogen leaching

Figure 3.1: A single realisation of $\mathbf{X}_t$ for $s = 3$ and $\beta_F = 200$.

The general trends of the Monte Carlo results displayed in Figure 3.2 agree with intuition. With increasing stocking rate and nitrogen fertilisation rate, we see increases in mean nitrogen concentration and nitrogen leaching. We also see a sharp decrease in mean pasture density as stocking rates increase. There is a consistent but less pronounced increase in mean pasture density as nitrogen fertilisation rates are increased. Unsurprising there is no variation in soil water storage across different stocking rates and nitrogen fertilisation rates.

Interestingly, for $\beta_F > 100$ we find that the mean nitrogen concentration is slightly greater for $s = 0$ than $s = 1$. This is surprising to see as we generally expect to see greater nitrogen when the stocking rate is increased. An explanation for this unexpected behaviour is found in pasture growth rates. For a stocking rate of $s = 0$, the pasture quickly grows to full capacity and then ceases to grow. From this point there is no nitrogen uptake by the pasture, as *the*



(a) Nitrogen concentration

(b) Soil water storage

(c) Pasture density

(d) Nitrogen leaching

Figure 3.2: Monte Carlo results over varied s and β_F parameters.

model assumes nitrogen uptake is proportional to pasture growth. Therefore nitrogen added via fertilisation decays slower than in cases where the grass is growing and up taking nitrogen, leading to our unexpected phenomena.

3.4.3 Validation With Existing Research

Table 3.2 compares the measurements from existing field research with those produced by Monte Carlo simulations of *the model*. We selected three collections of field data from Waikato dairy farms, which all recorded the stocking rate, nitrogen fertilisation rate and leached nitrogen quantity. When modelling the leaching rates, all parameters are held constant except for s and β_F. The same parameter set is used as defined in Table 3.1 with s and β_F being set to match the rates stated in the study of interest. We can see a general pattern of modelled leaching

Index	Period	s	β_F	Measured L_{rate}	Modelled L_{rate}	Reference
1	1993-96	3.3	0	40	28	[26]
2	1993-96	3.3	215	79	37	[26]
3	1993-96	3.3	413	150	50	[26]
4	1993-96	4.4	411	133	63	[26]
5	2004-05	2.90	117	27	28	[38]
6	2009-10	2.91	99	30	27	[38]
7	2014-15	2.93	99	30	28	[38]
8	2017-18	2.91	118	33	28	[38]
9	2011-14	3.2	150	54	32	[17]

Table 3.2: Comparison of measured nitrogen leaching and modelled leaching.

quantities being much more conservative than the measured quantities, particularly in Index 1, 2, 3, 4 and 9. The modelled results provide a good fit for Indexes 5, 6,7 and 8, although these indices are all from the same study.

3.5 Simplified Model

If we consider a simplified version of Eq. (3.3) and ignore the effects of soil water storage and pasture density, then some more analytical results can be generated. Let $X_t \in \mathbb{R}$ be the random process representing nitrogen concentration at time t in this simplified model (not to be confused with the bold type $\mathbf{X}_t \in \mathbb{R}^3$ previously defined). These alterations yield the following stochastic differential equation,

$$dX_t = f(X_t, t)dt + dY_t \tag{3.5}$$

where,

- $f(X_t, t)$ is the drift term, representing deterministic changes to X_t.
- Y_t is a CPP with rate λ and jump size distribution $g(z)$. dY_t represents changes in nitrogen concentration due to livestock urination events.

3.5.1 Probability Density Function

Let $\phi(x, t)$ be the probability density function of X_t as a function of x and t where x is the independent variable for nitrogen concentration. By the generalised Fokker-Planck equation derived by Denisov et al. [10], we are able to write a partial differential equation describing the evolution of $\phi(x,t)$ as,

$$\frac{\partial}{\partial t}\phi(x,t) = -\frac{\partial}{\partial x}f(x,t)\phi(x,t) - \lambda\phi(x,t) + \lambda\int_{-\infty}^{\infty}\phi(y,t)g(x-y)dy$$

3.5.2 Toy Case

Let us now analyse a simple case of Eq. (3.5) where the drift term models exponential decay of X_t at rate β. That is,

$$f(X_t,t) = -\beta X_t$$

We let the jump term Y_t model increases in X_t of fixed size α. Therefore, $g(z) = \delta(z-\alpha)$ where δ is the Dirac delta function. Therefore as Y_t has constant jump size of α we can write Y_t in terms of a PPP N_t of the same rate λ as such,

$$Y_t = \alpha N_t$$

Therefore, in this simple case we can write the SDE for X_t as,

$$dX_t = -\beta X_t dt + \alpha dN_t \tag{3.6}$$

The resulting PDE describing the evolution of $\phi(x,t)$ can be written as,

$$\frac{\partial}{\partial t}\phi(x,t) = \beta\frac{\partial}{\partial x}x\phi(x,t) + \lambda\phi(x-\alpha,t) - \lambda\phi(x,t) \tag{3.7}$$

Interestingly, Eq. (3.7) is exactly the same as Eq. (1.1) presented by Talyor et al. [35, 36]. Eq. (1.1) was derived as a density function for the distribution of nitrogen concentrations of a deterministic model across a whole paddock, whereas Eq. (3.7) was derived as a probability density function for the concentrations of a stochastic model at a single point in a paddock. To the best of our efforts, a closed-form solution to Eq. (3.7) could not be found, with numerical solutions being the best approach.

If we express the process N_t up to time T as a set of N_T arrival times, an explicit solution to Eq. (3.6) can be found. Let this set of jump times J be defined as,

$$J = \{\tau_1, \tau_2, \ldots, \tau_{N_T}\}$$

where $\tau_1 < \tau_2 < \cdots < \tau_{N_T}$. We can therefore write an analytical solution to Eq. (3.6) for the initial condition of X_0,

$$X_T = X_0 e^{-\beta T} + \alpha\sum_{i=1}^{N_T} e^{-\beta(T-\tau_i)}$$

Note that although this is a solution for X_t, it is still dependent on the arrival times in set J, which are randomly sampled according to N_t.

When using the simplified model defined in Eq. (3.6), then the parameter value β must be found empirically as it does not refer to any one quantifiable physical process. In addition, information regarding nitrogen leaching quantities is not immediately available, and a method of approximating the portion of exponential decay attributed to leaching must be developed.

Chapter 4

Numerical Methods

We will now cover the details of the numerical methods used to simulate realisations of $\mathbf{X}_t$ from Eq. (3.3). A simple jump-adapted Euler scheme is used. We expand upon the notes provided by Jum [21]. Let us recall Eq. (3.3) as,

$$d\mathbf{X}_t = \mathbf{f}(\mathbf{X}_t, t)dt + \mathbf{J}(\mathbf{X}_{t^-}, t^-)d\mathbf{Y}_t \tag{3.3}$$

We consider realisations of $\mathbf{X}_t$ over the time interval $t \in [0, T]$. Let us define $N(t)$ as a count of jump events across all three CPPs of the vector $\mathbf{Y}_t$ at time t. Let us write $\mathbf{Y}_t$ as a set of jump times J and a corresponding set of jump size vectors V over the interval $[0, T]$ where,

$$J = \{\tau_1, \tau_2, \ldots, \tau_{N(T)}\}$$

and,

$$V = \{\mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_{N(T)}\}$$

For $i \in \{1, 2, \ldots, N(T)\}$ the elements of each jump vector $\mathbf{V}_i \in \mathbb{R}^3$ represent the jump sizes for each of the three CPPs in $\mathbf{Y}_t$ at $t = \tau_i$. We introduce a time discretisation of $n + 1$ equidistant points over the interval $[0, T]$ as,

$$0 = \overline{\tau}_0 < \overline{\tau}_1 < \cdots < \overline{\tau}_n = T$$

We now define the jump-adapted time discretisation as,

$$\{t_0, t_1, \ldots, t_m\} = \{\overline{\tau}_0, \overline{\tau}_1, \ldots, \overline{\tau}_n\} \cup \{\tau_1, \tau_2, \ldots, \tau_{N(T)}\}$$

where $m+1$ is the number of elements in $\{\overline{\tau}_0, \overline{\tau}_1, \ldots, \overline{\tau}_n\} \cup \{\tau_1, \tau_2, \ldots, \tau_{N(T)}\}$. The indexing of the jump-adapted time discretisation corresponds to time ordering as such,

$$0 = t_0 < t_1 < \cdots < t_m = T$$

Let us define the jump-adapted time step Δt_i as,

$$\Delta t_i = t_{i+1} - t_i$$

We use the notation t_i^- and t_i^+ to represent the limit of t_i from the negative and positive directions, respectively. Given that the jump-adapted time discretisation includes all the times at which a jump occurs, there are no jumps within a given sub-interval (t_i, t_{i+1}), or equivalently $[t_i^+, t_{i+1}^-]$. Therefore the standard Euler step can be made within (t_i, t_{i+1}), as such,

$$\mathbf{X}(t_{i+1}^-) = \mathbf{X}(t_i^+) + \Delta t_i \mathbf{f}(\mathbf{X}(t_i^+), t_i^+)$$

Given that $\mathbf{X}(t)$ is continuous where there are no jumps, i.e. where $t \notin J$, we can write

$$\mathbf{X}(t_{i+1}^+) = \begin{cases} \mathbf{X}(t_{i+1}^-), & t_{i+1} \notin J \\ \mathbf{X}(t_{i+1}^-) + \mathbf{J}(\mathbf{X}(t_{i+1}^-), t_i)\mathbf{V}_{N(t_{i+1})}, & t_{i+1} \in J \end{cases}$$

The initial condition $\mathbf{X}_0$ gives,

$$\mathbf{X}(t_0^+) = \mathbf{X}_0$$

Although a simple jump-adapted Euler method has been introduced, we must still present a sensible method for numerically sampling the set of jump times J and the corresponding set of jump size vectors V. We have assumed constant Poisson rates λ_N, λ_W and λ_P for each of the elements $Y_N(t)$, $Y_W(t)$ and $Y_P(t)$ of $\mathbf{Y}_t$ respectively. We let $N_N(t)$, $N_W(t)$ and $N_P(t)$ be the corresponding PPPs with the same Poisson rates λ_N, λ_W and λ_P, respectively. It should be noted that the counting variable of all jump events $N(t)$ is not a PPP itself. Therefore, we must sample each PPP individually. By definition,

$$N_N(T) \sim \text{Pois}(\lambda_N T), \quad N_W(T) \sim \text{Pois}(\lambda_W T), \quad N_P(T) \sim \text{Pois}(\lambda_P T)$$

Therefore, we can simply sample the number of total events of each process over the interval $[0, T]$ from the distributions above using the MATLAB `poissrnd()` function (or equivalent).

Theorem 4.0.1 (Dekking et al., [9]). *Given that n events occur in a Poisson point process over the interval $[0, T]$, then the times of each of these events are independently distributed with a uniform distribution over $[0, T]$.*

Given that we have sampled particular values of $N_N(T)$, $N_W(T)$ and $N_P(T)$, Theorem 4.0.1 states we can sample the jump times from a uniform distribution. Let J_N, J_W and J_P be the sets of jump times for each process, respectively, with $N_N(T)$, $N_W(T)$ and $N_P(T)$ elements each, all elements being sampled from the uniform distribution $\mathcal{U}(0, T)$. We write J as,

$$J = J_N \cup J_W \cup J_P$$

As the jump size of each jump event is independent of time, we can sample the elements of V from the jump size distributions $g_N(z)$, $g_W(z)$ and $g_P(z)$ of Y_N, Y_W and Y_P respectively.

Remark. Althought not mentioned here, fertilisier application events are also included in J as they are modeled with the Dirac delta function.

Chapter 5

Discussion

5.1 Main Results

Numerous modelling assumptions are made throughout this dissertation. Clearly, a balance must be found between complexity and usability; however, given that results for the model must already be generated numerically, introducing additional complexities comes at a low cost. Several of the assumptions that could potentially be cause for concern are the same concerns raised by a review panel regarding the Overseer model as discussed in Section 1.2. *The model* uses a total nitrogen metric rather than multiple measures of various nitrogen compounds, only considers a single layer of soil/water and uses a single metric to quantify pasture density. Although these simplifications are necessary to produce a complete dissertation within the allocated timeframe, they come at the cost of modelling performance. *The model* does however consider discrete weather events through its stochastic modelling of rain events, this being a dynamic omitted from the Overseer model.

Another shortcoming of *the model* is the lack of any conservation laws of nitrogen. Clearly, total nitrogen must be conserved in a physical system, with total nitrogen inputs being equal to total nitrogen loss. To calculate conservation laws, additional metrics must be considered, such as nitrogen lost through milking and nitrogen gained via the supply of additional livestock feed.

The comparison of measured and modelled leaching rates presented in Section 3.4.3 was somewhat surprising. Several of the simulation parameters noted in Table 3.1 were derived rather crudely from the reference material. None of the parameters were calibrated. Instead, the parameters were sourced directly from reference material, which often gave relatively wide ranges. In addition, some parameters control very sensitive exponential or quadratic functions, potentially producing very large variance from small changes in parameter values. Therefore, it is somewhat surprising that the modelled leaching rates stated in Table 3.2 often reflected the true leaching rates. Even with some examples showing large errors, *the model*'s results were always within the correct magnitude. Although correctly modelling leaching rates within one magnitude is a very crude metric of success, it is still pleasing to see that *the model* presented in this dissertation is capable of producing reasonable outputs, given the complexity of the physical system at hand. This is particularly true when we consider that only two parameters (s and β_F) were adjusted to match the profile of each farm in Section 3.4.3.

All parameters, including rainfall event frequency and pasture growth rates, are constant. There is no consideration for seasonality within *the model*. Of course in reality we see large variations in such parameters across seasons. Rotational grazing is a common farming practice that could also be represented within *the model* by periodically changing parameters such as stocking rate.

5.2 Future Work

There are several avenues of future research to support the work presented in this dissertation.

With regards to the parameters used introduced in Section 3.2 and 3.3, later defined in Table 3.1, there is an opportunity to potentially improve the accuracy of modelled results by refining these parameter estimates. As previously mentioned, several parameters were crudely estimated from reference material in cases where little quantitative research had been conducted. When considering a particular farm, running a set of experiments to find accurate parameters specific to the soil type, grass type, and climate of the given farm would likely yield much-improved modelling results. For example, although all three studies presented in Table 3.2 are from the Waikato region, inevitably, there will still be variations in parameters between farms studied.

There is also an opportunity to adjust *the model* to use historical weather data, specific to a farm of interest. Particularly when performing numerical simulations, there could be great benefits in replacing the stochastic modelling of rainfall events with actual historical rainfall data. This could be useful in cases like Section 3.4.3 were we wish to compare measured and modelled leaching rates using field studies. Rather than generating weather events stochastically from long-term historic rainfall rates and frequencies, the actual rainfall data could be used from the same year in which the field data was recorded. During studies by Ledgard et al. over 1993 and 1994 in the Waikato, authors found that although the stocking rate and nitrogen fertilisation rate were almost identical between both years, the leaching rate in one specific paddock was over ten times greater in 1994 than 1993 (15 kg/ha in 1993 versus 204 kg/ha in 1994) [25]. The authors pointed out that weather conditions differed significantly between the two years, with 1993 seeing only 189 mm of drainage, whereas 1994 saw 551 mm of drainage, a potential reason for the varied leaching results. This would also allow us to understand if *the model* can give accurate leaching results for weather patterns which do not do not follow a conventional distribution.

More rigorous validation and calibration are needed against recorded field data. In Section

3.4.3 we compared research which presented annual leaching rates for several dairy farms of varying characteristics against modelled leaching. Although this section made a good effort to begin validating *the model*, there are still opportunities for additional work to be done. There was no validation of the components of $\mathbf{X}_t$ over time, or the distribution of leaching rates. We would be more confident in *the model*'s estimates of leaching rates if we had validated the intermediate steps required to calculate leaching rates.

An alternative approach to improving the modelling performance could be found in adapting a sophisticated model such as the Daisy model [19], the DNDC model [27], or the APSIM model [22] to include urine patch dynamics.

Chapter 6

Conclusions

We have proposed, justified and evaluated a system of SDEs to model nitrogen leaching in dairy farming. *The model* considers the evolution of nitrogen concentrations, soil water storage and pasture density over time, with considerations for the stochastic changes in all three quantities. Particular focus is placed on the dynamics of high nitrogen concentration urine patches and the resulting effect on nitrogen leaching rates. After producing numerical Monte Carlo simulation results we found that although *the model* produced qualitatively correct results for nitrogen leaching, errors are still large, pointing towards the need for parameter calibration and/or additional research.

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